ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Chemical Bonding

English Medium

EXERCISE # O-1

WEAK FORCES

1. **Statement-1 :** The melting point of noble gases increases as its atomic mass increases.
Statement-2 : Instantaneous dipole induced dipole attraction increases with increase in atomic mass of noble gases.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CB0001

2. The critical temperature of water is higher than that of O_2 because the H_2O molecule has :
(A) fewer electrons than O_2 (B) two ionic bonds
(C) V–shape (D) dipole moment

CB0002

3. Which of the following boiling point order is correct -
(A) $He > T_2 > D_2$ (B) $He < T_2 < D_2$ (C) $T_2 > He > D_2$ (D) $He < D_2 < T_2$

CB0003

4. Which is the incorrect match for the energy distance function for following interaction -
(A) Debye force : r^{-6} (B) Ion-induced dipole interaction : r^{-2}
(C) London force : r^{-6} (D) Keesom force : r^{-3}

CB0004

5. Identify the incorrect order of boilng point in the following pair.
(A) $B(OH)_3 < B(OCH_3)_3$ (B) $NF_3 < N(CH_3)_3$
(C) $BF_3 < B(CH_3)_3$ (D) $C_2H_6 < C_2F_6$

CB0005

MULTICENTERED BOND

6. The type of overlap in the bridge bond existing in $Al_2(CH_3)_6$ is :-
(A) $sp^3–sp^3d–sp^3$ (B) $sp^3–sp^2–sp^3$ (C) $sp^3–s–sp^3$ (D) $sp^3–sp^3–sp^3$

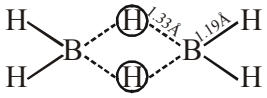
CB0006

7. Which one of the following statement is not true regarding diborane?
(A) It has two bridging hydrogens and four perpendicular to the rest.
(B) When methylated, the product is $Me_4B_2H_2$
(C) The bridging hydrogens are in a plane perpendicular to the rest.
(D) All the B–H bond distances are equal.

CB0007

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8. The molecular shape of diborane, is shown:



Consider the following statements for diborane :

(1) Boron is approximately sp^3 hybridised

(2) B–H–B angle is 180°

(3) There are two terminal B–H bonds for each boron atom

(4) There are only 12 bonding electrons available

Of these statements:

(A) 1, 3 and 4 are correct (B) 1, 2 and 3 are correct

(C) 2, 3 and 4 are correct (D) 1, 2 and 4 are correct

CB0008

ODD ELECTRON SPECIES

9. Hybridisation related to NO_2 molecule is -

(A) sp^3 (B) sp (C) sp^3d (D) sp^2

CB0009

10. In which of the following processes, the magnetic behaviour of the species is changed :-

(A) $2\dot{C}H_3 \longrightarrow C_2H_6$ (B) $2NO_2 \longrightarrow N_2O_4$

(C) $2ClO_3 \longrightarrow Cl_2O_6$ (D) All of these

CB0010

HYDROLYSIS

11. Which of the following statement is correct ?

(A) BCl_3 is not hydrolysed while $SiCl_4$ can be hydrolysed

(B) CCl_4 is hydrolysed under ordinary condition

(C) XeF_2 produces $Xe(OH)_2$ on hydrolysis

(D) hydrolysis of XeF_2 is a redox reaction

CB0011

12. **Statement-1 :** Between $SiCl_4$ and CCl_4 only $SiCl_4$ reacts with water at room temperature.

Statement-2 : $SiCl_4$ is ionic and CCl_4 is covalent.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

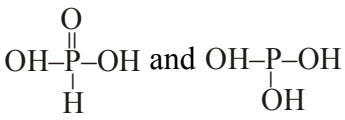
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

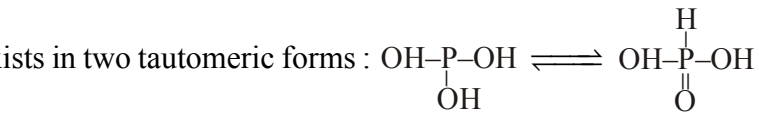
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0012

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13. **Statement-1 :** PCl_3 on hydrolysis gives  and

Statement-2 : H_3PO_3 exists in two tautomeric forms : 

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0013

INERT PAIR EFFECT

14. **Statement-1 :** Boron does not show univalent nature but unipositive nature of thallium is quite stable.

Statement-2 : Inert pair effect predominates in thallium.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0014

15. **Statement-1 :** PbI_4 doesn't exist and converts into PbI_2 and I_2 spontaneously at room temperature but $PbCl_4$ needs heating to convert into $PbCl_2$ and Cl_2.

Statement-2 : Pb^{2+} is more stable than Pb^{4+} due to inert pair effect.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0015

IONIC COMPOUND

16. The dissolution of ionic compounds involves :

(A) Evolution of heat (B) Weakening of attractive forces

(C) Dissociation into ions (D) All of these

CB0016

17. Select correct order out of given options :

(A) $BeCO_3 < BaCO_3$: Covalent character (B) BeO > SrO : lattice energy

(C) $Be^{2+} < Li^+$: Hydration energy (D) $Be^{2+}(aq.) > Li^+(aq.)$: Ionic mobility

CB0017

18. The polarizibility of the following ions is/are in the order of

(A) $I^- > Br^- > Cl^- > F^-$ (B) $I^- > Br^- > F^- > C1^-$

(C) $I^- < Br^- < Cl^- < F^-$ (D) $I^- < Br^- < F^- < Cl^-$

CB0018

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19. Which of the following equilibria would have the highest value of K_p at a common temperature?

(A) $MgCO_3 \Leftrightarrow MgO + CO_2$ (B) $CaCO_3 \Leftrightarrow CaO + CO_2$

(C) $SrCO_3 \Leftrightarrow SrO + CO_2$ (D) $BaCO_3 \Leftrightarrow BaO + CO_2$

CB0019

20. Which of the following set of characteristics lead to the increase in solubility of ionic substances?

(A) High dipole moment, strong attraction by an ion towards solvent and large solvation energy

(B) Low dipole moment, weak attraction by an ion towards solvent and high solvation energy

(C) High dipole moment, strong attraction by an ion towards solvent and low solvation energy

(D) High dipole moment, weak attraction by an ion towards solvent and large solvation energy

CB0020

21. The solubility of anhydrous $AlCl_3$ and hydrated $AlCl_3$ in diethyl ether are S_1 and S_2 respectively. Then

(A) $S_1 = S_2$ (B) $S_1 > S_2$ (C) $S_1 < S_2$ (D) $S_1 < S_2$ but not $S_1 = S_2$

CB0021

22. **Statement-1 :** Among alkali metal cations, $Li^+(aq.)$ has highest electrical conductance.

Statement-2 : $Li^+(aq.)$ is largest alkali metal cation because of greater degree of hydration.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0022

23. **Statement-1 :** $Al(OH)_3$ is amphoteric in nature.

Statement-2 : Al–O and O–H bonds can be broken with equal ease in $Al(OH)_3$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0023

MOLECULAR ORBITAL THEORY

24. The bond energy order of He_2^+ and HeH^+ is

(A) $He_2^+ > HeH^+$ (B) $HeH^+ = He_2^+$ (C) $HeH^+ > He_2^+$ (D) Can't be predicted

CB0024

25. Among KO_2, AlO_2^-, BaO_2 and NO_2^+ unpaired electron is present in :

(A) KO_2 only (B) NO_2^+ and BaO_2 (C) KO_2 and AlO_2^- (D) BaO_2 only

CB0025

26. During the formation of a molecular orbital from atomic orbitals, probability of electron density is

(A) minimum in the nodal plane (B) maximum in the nodal plane

(C) zero in the nodal plane (D) zero on the surface of the lobe

CB0026

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27. Pick out the **incorrect** statement ?

(A) N_2 has greater dissociation energy than N_2^+

(B) O_2 has lower dissociation energy than O_2^+

(C) Bond length in N_2^+ is less than N_2

(D) Bond length in NO^+ is less than in NO

CB0027

28. A simplified application of MO theory to the hypothetical molecule 'OF' would give its bond order as :

(A) 2 (B) 1.5 (C) 1.0 (D) 0.5

CB0028

29. Which of the following is true ?

(A) With increasing Bond order , Bond length decreases & Bond energy increases

(B) With increasing Bond order , Bond length increases & Bond energy decreases

(C) With increasing Bond order , Bond length decreases & Bond energy decreases

(D) With increasing Bond order , Bond length increases & Bond energy increases

CB0029

30. Which of the following has fractional bond order :

(A) O_2^{2+} (B) O_2^{2-} (C) F_2^{2-} (D) H_2^-

CB0030

31. **Statement-1:** H_2 molecule is more stable than He–H molecule.

Statement-2 : The antibonding electron in He–H molecule decreases the bond order and there by the stability.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0031

32. **Statement-1 :** Super oxide ion is paramagnetic whereas peroxide ion is diamagnetic.

Statement-2 : Super oxide ion has one unpaired electron whereas peroxide ion has no unpaired electron.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CB0032

33. **Statement-1:** π_b and π^* orbitals obtained from 2p orbital are lying in the same plane.

Statement-2 : Bonding M.O's are formed by constructive interference while antibonding M.O's are formed by destructive interference.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CB0033**

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EXERCISE # O-2

WEAK FORCES

1. Which of the following factors are responsible for origination of vander Waals forces ?
(A) Instantaneous dipole-induced dipole interaction
(B) Dipole-induced dipole interaction
(C) Dipole-dipole interaction
(D) Size of molecule

CB0034

2. Which of the following are true ?
(A) Vander Waals forces are responsible for the formation of molecular crystals
(B) Branching lowers the boiling points of isomeric organic compounds due to decrease in Vander Waals forces of attraction
(C) In graphite, vander Waals forces act between the carbon layers
(D) In diamond, vander Waals forces act between the carbon layers

CB0035

MULTICENTERED BOND

3. No X–X bond exists in which of the following compounds having general form of X_2H_6 ?
(A) B_2H_6 (B) C_2H_6 (C) Al_2H_6 (D) Si_2H_6

CB0036

4. Three centre - two electron bonds exist in :
(A) B_2H_6 (B) $Al_2(CH_3)_6$ (C) $BeH_2(s)$ (D) $BeCl_2(s)$

CB0037

5. Select correct statement about B_2H_6
(A) Bridging groups are electron-deficient with 12 valence electrons
(B) It has 2c – 2e B–H bonds
(C) It has 3c – 2e B–H–B bonds
(D) It has 3c – 4e B–H–B bonds

CB0038

ODD ELECTRON SPECIES

6. Select correct about NO_2 :
(A) It is odd electron specie
(B) N–O bond order = 1.5
(C) Paramagnatic specie
(D) Isoelectronic with CO_2

CB0039

7. The number of specie(s) which are not perfectly planar.
(A) $\dot{C}H_3$ (B) $\dot{C}F_3$ (C) $\dot{C}HF_2$ (D) $\dot{C}H_2F$

CB0040

8. Which of the following statement is **CORRECT** :-
(A) The free electron of ClO_3 molecule is present in d-orbital of Cl-atom
(B) The free electron of $\dot{C}F_3$ is present in sp^3 hybrid orbital
(C) NO is polar
(D) The free electron of ClO_2 molecule is present in d-orbital of Cl-atom

CB0041

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9. Which of the following statement(s) is / are **INCORRECT** for

$\dot{C}H_3$ = X and $\dot{C}F_3$ = Y

(A) When X dimerises bond angle decreases

(B) When X dimerises bond angle increases

(C) In X–Y molecule C–C bond length less than that in Y–Y molecule

(D) Bond angle in Y is less than X

CB0042

HYDROLYSIS

10. Which of the following halide(s) cannot be hydrolysed at room temperature

(A) SeF_6 (B) SF_6 (C) CCl_4 (D) NF_3

CB0043

11. Which of the following statement is/are **INCORRECT** ?

(A) P_4S_{10} gives rise to H_2S gas on hydrolysis

(B) PCl_5 produces $POCl_3$ on partial hydrolysis

(C) H_2SO_5 gives rise to H_2SO_3 on hydrolysis

(D) d-orbital participates in the hydrolysis of SF_6 at room temperature

CB0044

INERT PAIR EFFECT

12. Which of the following have (18 + 2) electron configuration ?

(A) Pb^{2+} (B) Cd^{2+} (C) Bi^{3+} (D) SO_4^{2-}

CB0045

13. Which of following stability order is/are correct due to inert pair effect.

(A) $Hg > Hg^{2+}$ (B) $Bi^{3+} < Bi^{5+}$ (C) $Pb^{2+} > Pb^{4+}$ (D) $Fe^{2+} < Fe^{3+}$

CB0046

IONIC COMPOUND

14. Choose the correct order(s) for the given properties.

(A) $MgSO_4 < SrSO_4 < BaSO_4$: Thermal stability order

(B) $BeC_2O_4 < CaC_2O_4 < BaC_2O_4$: Solubility in water

(C) $LiCl > NaCl > KCl$: Melting point order

(D) $BeF_2 > CaF_2 > SrF_2$: Covalent character order

CB0047

15. Polarization may be called the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is/are incorrect :

(A) Minimum polarization is brought about by a cation of low radius

(B) A large cation is likely to bring about a large degree of polarization

(C) Maximum polarization is brought about by a cation of high charge

(D) A small anion is likely to undergo a large degree of polarization

CB0048

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16. Most ionic compounds have :

(A) high melting points and low boiling points

(B) high melting points and nondirectional bonds

(C) high solubilities in polar solvents and low solubilities in nonpolar solvents

(D) three-dimentional network structures, and are good conductors of electricity in the molten state

CB0049

17. Choose the correct order for the given properties.

(A) $NaF < MgF_2 < AlF_3$: covalent character order.

(B) $NaF < MgF_2 < AlF_3$: melting point order

(C) $NaF < MgF_2 < AlF_3$: lattice energy order

(D) $NaF > MgF_2 > AlF_3$: order of polarising power of cation.

CB0050

MOLECULAR ORBITAL THEORY

18. Which of the following have identical bond order ?

(A) O_2^{2+} (B) NO^+ (C) CN^- (D) CN^+

CB0051

19. Which of the following statement is/are correct

(A) The peroxide ion has a bond order of 1 while the oxygen molecule has a bond order of 2

(B) The peroxide ion has a weaker bond than the dioxygen molecule has.

(C) The peroxide ion as well as the dioxygen molecules are paramagnetic

(D) The bond length of the peroxide ion is greater than that of the dioxygen molecule

CB0052

20. Which of the following statements are true for these given species : N_2, CO, CN^- and NO^+.

(A) All species are paramagnetic (B) The species are isoelectronic

(C) All the species have dipole moment (D) All the species are linear

CB0053

21. Which of the following have unpaired electron(s)

(A) O_2^+ (B) O_2^- (C) NO (D) H_2^+

CB0054

22. Which of the following is/are paramagnetic ?

(A) B_2 (B) O_2 (C) N_2 (D) He_2

CB0055

23. Which of the following species have a bond order of 3 ?

(A) CO (B) CN^- (C) NO^+ (D) O_2^+

CB0056

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24. Which of the following is/are correct ?

(A) During N_2^+ formation, one electron is removed from the bonding molecular orbitals

(B) During O_2^+ formation, one electron is removed from the antibonding molecular orbitals

(C) During O_2^- formation, one electron is added to the bonding molecular orbitals

(D) During CN^- formation, one electron is added to the bonding molecular orbitals

CB0057

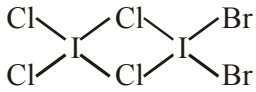
MISCELLANEOUS

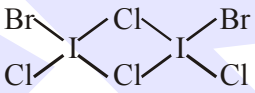
25. Rotation around the bond (between the underlined atoms) is restricted in :

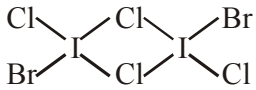
(A) $\underline{C}_2H_4$ (B) $H_2\underline{O}_2$ (C) $\underline{C}_2H_2$ (D) $\underline{C}_2H_6$

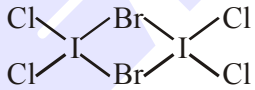
CB0058

26. The experimental result says that two iodine atoms are in different environments and predict the all possible incorrect arrangement for $I_2Cl_4Br_2$.

(A) 

(B) 

(C) 

(D) 

CB0059

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EXERCISE # S-1

1. Find the number of angles less than 120° in PH_2F_3.

CB0060

2. Find the number of molecules which are not having 3c–2e bond from the following.

$Al_2(CH_3)_6$, Si_2H_6 , B_2H_6 , C_2H_6 , Si_2Cl_6 , Al_2Cl_6

CB0061

3. Find the ratio of π-electrons in the C_2-molecule with that of B_2 molecule according to M.O.T.

CB0062

4. The total number of bonding and antibonding electrons in O_2^+ are "....." and "....." respectively.

[If the answer is 14 and 7, then represent as 147]

CB0063

5. Find the total number of following molecule(s) which have all bond lengths are same.

XeF_4, SF_4, SH_2, NO_3^-, SiF_4, ClF_3, PF_2Cl_3, XeO_3F_2

CB0064

6. Among the following total number of planar molecules is / are

Cl_2O , $P(CH_3)_3$, $N(CH_3)_3$,ClO_2 , $\dot{C}H_3$, NCl_3

CB0065

7. How many compound(s) gives diprotic acid on hydrolysis ?

SO_2Cl_2, $SOCl_2$, $POCl_3$, XeF_2, N_2O_5, P_4O_6, SF_4, BCl_3

CB0066

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EXERCISE # S-2

Comprehension # 1 (1 to 3 Questions)

B is the first element of III[rd] group. It forms a number of electron deficient halides and hydrides. Among the hydrides diborane is an important compound.

1. Which of the following halide is the strongest Lewis acid?

(A) BF_3 (B) BCl_3 (C) BBr_3 (D) BI_3

CB0067

2. Which of the following compounds has 2pπ - 2pπ bond ?

(A) BF_3 (B) BCl_3 (C) BBr_3 (D) BI_3

CB0067

3. In B_2H_6 number of 3c – 2e bonds is/are

(A) 1 (B) 2 (C) 3 (D) None

CB0067

Comprehension # 2 (4 to 6 Questions)

Molecular orbital theory is based on linear combination of atomic orbitals (LCAO). According to LCAO when respective atomic orbitals of the atoms interact, they undergoes constructive and destructive interference giving two types of molecular orbital i.e. bonding and antibonding molecular orbitals respectively.

4. Which of the following specie is paramagnetic ?

(A) NO^- (B) O_2^{2-} (C) CN^- (D) CO

CB0068

5. Bond order of Be_2 is :

(A) 1 (B) 2 (C) 3 (D) 0

CB0068

6. Number of anti bonding electrons in N_2 is :

(A) 4 (B) 10 (C) 12 (D) 14

CB0068

Comprehension # 3 (7 to 8 Questions)

Polarisation of anion in ionic compounds play an important role to influnce the various physical and chemical properties of ionic compound.

7. Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are :

(A) LiCl and RbCl (B) RbCl and $BeCl_2$ (C) RbCl and $MgCl_2$ (D) $MgCl_2$ and $BeCl_2$

CB0069

8. Compound with maximum ionic character is formed from :

(A) Na and Cl (B) Cs and F (C) Cs and I (D) Na and F

CB0069

note\06\606C-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\Word #\Chemical Bonding-Co-chem & Metallurgy\(OC)\Eng\01-CB\02 Exercise.p65

Comprehension # 4 (9 to 10 Questions)

"Hydrolysis is defined as the reaction of water with any susbtance"

9. Choose the correct order of ease of hydrolysis -

(A) $MgCl_2 > AlCl_3$ (B) $SF_6 < SeF_6$ (C) $SnCl_2 > SnCl_4$ (D) None

CB0070

10. Which of the following oxyacids are formed during the stepwise hydrolysis of P_4O_{10}

(A) tetrametaphosphoric acid (B) tetrapolyphosphoric acid

(C) pyrophosphoric acid (D) All of these

CB0070

MATCHING LIST

11. Match list I with list II and select the correct answer:

List I (species)	**List II (O-N-O angle)**
(P) NO_2^+	(1) 180°
(Q) NO_2	(2) 134°
(R) NO_2^-	(3) 120°
(S) NO_3^-	(4) 115°
	(5) 109°

	P	**Q**	**R**	**S**		**P**	**Q**	**R**	**S**
(A)	5	4	3	2	(B)	5	2	4	3
(C)	1	2	4	3	(D)	1	4	3	2

CB0071

12. Match list I with list II and select the correct answer:

List–I (Molecule / Species)	**List–II (Unpaired electron resides in)**
(P) NO_2	(1) d-orbital
(Q) ClO_2	(2) sp^2-orbital
(R) ClO_3	(3) sp^3-orbital
(S) $^{\cdot}CH_3$	(4) p-orbital

Code :

	P	**Q**	**R**	**S**		**P**	**Q**	**R**	**S**
(A)	2	4	1	3	(B)	2	1	3	4
(C)	1	4	2	3	(D)	3	1	2	4

CB0072

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13. Match list I with list II and select the correct answer:

List–I (Process)	**List–II (Operating intraction involved)**
(P) Clathrate compound of Xe in ice	(1) Ion - Dipole
(Q) Liquation of Xe gas	(2) Dipole - Dipole
(R) Liquation of HCl gas	(3) Dipole - Induced dipole
(S) Hydration of Na^+	(4) London forces

Code :

	P	Q	R	S
(A)	1	3	4	2
(C)	3	4	2	1

	P	Q	R	S
(B)	3	4	1	2
(D)	1	3	2	4

CB0073

MATRIX MATCH

14. Match the column

Column-I (Compounds given)	**Column-II (Characteristics associated with given compounds)**
(A) $BeCl_2$	(P) Undergoes partial hydrolysis
(B) SiF_4	(Q) All possible bond angles are identical
(C) SO_2Cl_2	(R) Hydrolysed product of the attacking site is electron deficient and finally produces polymerised product
(D) BF_3	(S) Maximum number of atoms present in one plane is three

CB0074

15. Match the column

Column-I (Molecules)	**Column-I (Characteristics given among the molecules of column-I)**
(A) CH_4	(P) Molecule is having perfect tetrahedral shape
(B) CH_2F_2	(Q) C-F bond has maximum p-character
(C) CHF_3	(R) C–H bond has maximum s-character
(D) CF_4	(S) Molecule is having maximum number of equal angles
	(T) Molecule has lowest bond angle

CB0075

16. Match the column

Column-I Processes	**Column-II Correct characteristics**
(A) $N_2^+ \longrightarrow N_2$	(P) Magnetic moment gets changed
(B) $Zn^{2+} \longrightarrow Zn$	(Q) The process is associated with two electronic change
(C) $O_2^{2-} \longrightarrow 2O^{2-}$	(R) Magnetic behaviour gets changed
(D) $C_2^{2-} \longrightarrow C_2$	(S) Electron(s) associated in the process enter(s) into π^*_{2p} orbital
	(T) Electron(s) associated in the process involve(s) σ_{2p} orbital

CB0076

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Answer Q.17 to Q.19 by appropriately matching the information given in the three columns of the following table

Molecular Orbital	Number of nodal plane	Symmetry of Molecular Orbital
(1) σ_s	(P) 3	(I) BMO, gerade
(2) π_p	(Q) 2	(II) ABMO, ungerade
(3) $\sigma^*_{p_z}$	(R) 1	(III) BMO, ungerade
(4) δ^*	(S) 0	(IV) ABMO, gerade

17. Which of the following matching is **INCORRECT.**

(A) (1), S, I (B) (2), R, I (C) (3), R, II (D) (4), P, II

CB0077

18. How many orbitals are occupied with set (1)(S)(I) in F_2

(A) 0 (B) 1 (C) 2 (D) 3

CB0077

19. If z axis is the molecular axis then **CORRECT** matching for $d_{xy} + d_{xy}$ orbital is 4, P, II for the substraction of wave function.

Which of the following matching is **CORRECT** for the addition of wave function for same combination of orbitals.

(A) P, IV (B) P, III (C) Q, II (D) Q, I

CB0077

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EXERCISE # JEE-MAIN

1. Among the following the maximum covalent character is shown by the compound :- **[AIEEE-2011]**

(1) $AlCl_3$ (2) $MgCl_2$ (3) $FeCl_2$ (4) $SnCl_2$

CB0091

2. Which one of the following molecules is expected to exhibit diamagnetic behaviour ?

(1) C_2 (2) N_2 (3) O_2 (4) S_2 **[AIEEE-2013]**

CB0092

3. In which of the following pairs of molecules/ions, both the species are not likely to exist ?

(1) H_2^+, He_2^{2-} (2) H_2^-, He_2^{2-} (3) H_2^{2+}, He_2 (4) H_2^-, He_2^{2+} **[JEE-M-2013]**

CB0093

4. Stability of the species Li_2, Li_2^- and Li_2^+ increases in the order of :- **[JEE-M-2013]**

(1) $Li_2 < Li_2^+ < Li_2^-$ (2) $Li_2^- < Li_2^+ < Li_2$ (3) $Li_2 < Li_2^- < Li_2^+$ (4) $Li_2^- < Li_2 < Li_2^+$

CB0094

5. Which one of the following properties is **not** shown by NO ? **[JEE-M-2014]**

(1) It combines with oxygen to form nitrogen dioxide

(2) It's bond order is 2.5

(3) It is diamagnetic in gaseous state

(4) It is a neutral oxide

CB0095

6. The correct order of thermal stability of hydroxides is : **[JEE-M-2015 (on line)]**

(1) $Ba(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Mg(OH)_2$

(2) $Mg(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Ba(OH)_2$

(3) $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

(4) $Ba(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Mg(OH)_2$

CB0096

7. Which of the alkaline earth metal halides given below is essentially covalent in nature :- **[JEE-M-2015 (on line)]**

(1) $SrCl_2$ (2) $CaCl_2$ (3) $BeCl_2$ (4) $MgCl_2$

CB0097

8. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ? **[JEE-M-2015]**

(1) $BaSO_4$ (2) $SrSO_4$ (3) $CaSO_4$ (4) $BeSO_4$

CB0098

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9. The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is :- **[JEE-M-2015]**

(1) London force (2) Hydrogen bond

(3) ion-ion interaction (4) ion-dipole interaction

CB0099

10. Which one has the highest boiling point ? **[JEE-M-2015]**

(1) Kr (2) Xe (3) He (4) Ne

CB0100

11. Which intermolecular force is most responsible in allowing xenon gas to liquefy? **[JEE (MAIN) ONLINE 2016]**

(1) Ionic

(2) Instantaneous dipole- induced dipole

(3) Dipole - dipole

(4) Ion - dipole

CB0101

12. The bond angle H–X–H is the greatest in the compound : **[JEE (MAIN) ONLINE 2016]**

(1) NH_3 (2) H_2O (3) PH_3 (4) CH_4

CB0102

13. Which of the following species is not paramagnetic :- **[JEE-MAINS-2017]**

(1) NO (2) CO (3) O_2 (4) B_2

CB0103

14. Which of the following is paramagnetic ? **[JEE-MAINS-2017 (On-line)]**

(1) CO (2) O_2^{2-} (3) NO^+ (4) B_2

CB0104

15. sp^3d^2 hybridization is not displayed by : **[JEE-MAINS-2017 (On-line)]**

(1) $[CrF_6]^{3-}$ (2) BrF_5 (3) PF_5 (4) SF_6

CB0105

16. The number of S=O and S–OH bonds present in peroxodisulphuric acid and pyrosulphuric acid respectively are : **[JEE-MAINS-2017 (On-line)]**

(1) (2 and 4) and (2 and 4) (2) (4 and 2) and (2 and 4)

(3) (2 and 2) and (2 and 2) (4) (4 and 2) and (4 and 2)

CB0106

17. The correct sequence of decreasing number of π-bonds in the structures of H_2SO_3, H_2SO_4 and $H_2S_2O_7$ is:- **[JEE-MAINS-2017 (On-line)]**

(1) $H_2S_2O_7 > H_2SO_4 > H_2SO_3$ (2) $H_2SO_3 > H_2SO_4 > H_2S_2O_7$

(3) $H_2S_2O_7 > H_2SO_3 > H_2SO_4$ (4) $H_2SO_4 > H_2S_2O_7 > H_2SO_3$

CB0107

18. The increasing order of the boiling points for the following compounds is:-

[JEE-MAINS-2017 (On-line)]

(I) C_2H_5OH (II) C_2H_5Cl (III) $C_2H_5CH_3$ (IV) $C_2H_5OCH_3$

(1) (III) < (II) < (I) < (IV) (2) (II) < (III) < (IV) < (I)

(3) (IV) < (III) < (I) < (II) (4) (III) < (IV) < (II) < (I) **CB0108**

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19. The number of P–OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($H_4P_2O_7$) respectively are :- **[JEE-MAINS-2017 (On-line)]**

(1) five and four (2) five and five (3) four and five (4) four and four

CB0109

20. The group having triangular planar structures is :- **[JEE-MAINS-2017 (On-line)]**

(1) CO_3^{2-}, NO_3^-, SO_3 (2) NCl_3, BCl_3, SO_3 (3) NH_3, SO_3, CO_3^{2-} (4) BF_3, NF_3, CO_3^{2-}

CB0110

21. Which of the following are Lewis acids ? **[JEE-MAINS-2018]**

(1) $AlCl_3$ and $SiCl_4$ (2) PH_3 and $SiCl_4$ (3) BCl_3 and $AlCl_3$ (4) PH_3 and BCl_3

CB0111

22. According to molecular orbital theory, which of the following will not be a viable molecule ? **[JEE-MAINS-2018]**

(1) He_2^+ (2) H_2^- (3) H_2^{2-} (4) He_2^{2+}

CB0112

23. Which of the following compounds contain(s) no covalent bond(s) ? **[JEE-MAINS-2018]**

KCl , PH_3 , O_2 , B_2H_6 , H_2SO_4

(1) KCl, H_2SO_4 (2) KCl (3) KCl, B_2H_6 (4) KCl, B_2H_6, PH_3

CB0113

24. In KO_2, the nature of oxygen species and the oxidation state of oxygen atom are, respectively **[JEE-MAINS-2018 (Online]**

(1) Superoxide and –1/2 (2) Oxide and –2

(3) Peroxide and –1/2 (4) Superoxide and –1

CB0114

25. Which of the following best describes the diagram below of a moleuclar orbital ? **[JEE-MAINS-2018 (Online)]**



(1) An antibonding π orbital (2) An antibonding σ orbital

(3) A non-bonding orbital (4) A bonding π orbital

CB0115

26. In the molecular orbital diagram for the molecular ion, N_2^+, the number of electrons in the σ_{2p} molecular orbital is :- **[JEE-MAINS-2018 (Online)]**

(1) 3 (2) 1 (3) 0 (4) 2

CB0116

27. Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y' Compounds 'X' and 'Y' and the oxidation state of Xe are respectively : **[JEE-MAINS-2018 (Online)]**

(1) XeO_2F_2(+6) and XeO_2(+4) (2) $XeOF_4$(+6) and XeO_2F_2(+6)

(3) $XeOF_4$(+6) and XeO_3(+6) (4) XeO_2(+4) and XeO_3(+6)

CB0117

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28. Which of the following is a lewis acid ? **[JEE-MAINS-2018 (Online)]**

(1) NaH (2) NF_3 (3) PH_3 (4) $B(CH_3)_3$

CB0118

29. A group 13 element 'X' reacts with chlorine gas to produce a compound XCl_3. XCl_3 is electron deficient and easily reacts with NH_3 to form $Cl_3X \leftarrow NH_3$ adduct; however, XCl_3 does not dimerize. X is :- **[JEE-MAINS-2018 (Online)]**

(1) Ga (2) Al (3) In (4) B

CB0119

30. Which of the following conversions involves change in both shape and hybridisation ? **[JEE-MAINS-2018 (Online)]**

(1) $BF_3 \rightarrow BF_4^-$ (2) $H_2O \rightarrow H_3O^+$ (3) $CH_4 \rightarrow C_2H_6$ (4) $NH_3 \rightarrow NH_4^+$

CB0120

31. Among the sulphates of alkaline earth metals, the solubilities of $BeSO_4$ and $MgSO_4$ in water, respectively, are: **[JEE-MAINS-2020]**

(1) high and high (2) poor and poor (3) high and poor (4) poor and high

CB0168

32. The structure of PCl_5 in the solid state is **[JEE-MAINS-2020]**

(1) square pyramidal

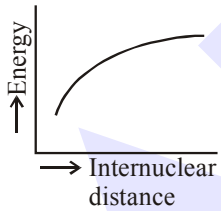
(2) tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$

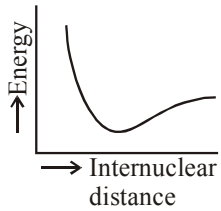
(3) square planar $[PCl_4]^+$ and octahedral $[PCl_6]^-$

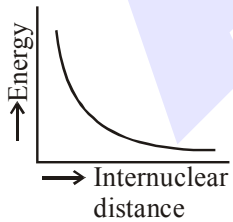
(4) trigonal bipyramidal

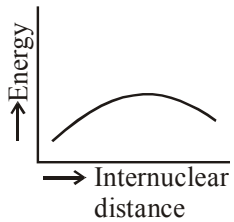
CB0169

33. The potential energy curve for the H_2 molecule as a function of internuclear distance is : **[JEE-MAINS-2020]**

(1)


(2)


(3)


(4)


CB0170

34. The compound that has the largest H–M–H bond angle (M=N, O, S, C), is : **[JEE-MAINS-2020]**

(1) H_2O (2) CH_4 (3) NH_3 (4) H_2S

CB0171

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35. Match the type of interaction in Column A with the distance dependence of their interaction energy in Column B : **[JEE-MAINS-2020]**

A	B
(I) iron - ion	(a) $\frac{1}{r}$
(II) dipole - dipole	(b) $\frac{1}{r^2}$
(III) London dispersion	(c) $\frac{1}{r^3}$
	(d) $\frac{1}{r^6}$

(1) (I)-(a), (II)-(b), (III)-(c) (2) (I)-(a), (II)-(c), (III)-(d)
(3) (I)-(a), (II)-(b), (III)-(d) (4) (I)-(b), (II)-(d), (III)-(c)

CB0172

36. If the boiling point of H_2O is 373 K, the boiling point of H_2S will be : **[JEE-MAINS-2020]**
(1) Greater than 300 K but less than 373 K (2) Less than 300 K
(3) Equal to 373 K (4) More than 373 K

CB0173

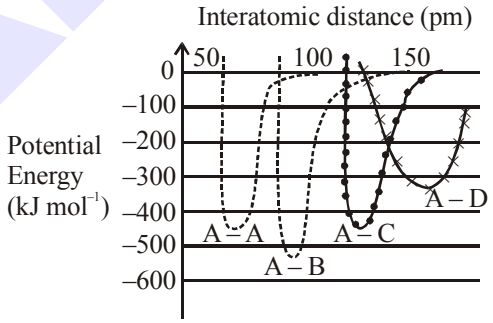
37. In a molecule of pyrophosphoric acid, the number of P–OH, P=O and P–O–P bonds/ moiety(ies) respectivey are : **[JEE-MAINS-2020]**
(1) 3, 3 and 3 (2) 2, 4 and 1 (3) 4, 2 and 0 (4) 4, 2 and 1

CB0174

38. Of the species, NO, NO^+, NO^{2+}, NO^-, the one with minimum bond strength is : **[JEE-MAINS-2020]**
(1) NO^{2+} (2) NO^+ (3) NO (4) NO^-

CB0175

39. The intermolecular potential energy for the molecules A, B, C and D given below suggests that : **[JEE-MAINS-2020]**



(1) D is more electronegative than other atoms
(2) A-D has the shortest bond length
(3) A-B has the stiffest bond
(4) A–A has the largest bond enthalpy

CB0176

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40. If AB_4 molecule is a polar molecule, a possible geometry of AB_4 is : **[JEE-MAINS-2020]**

(1) Square pyramidal (2) Tetrahedral

(3) Square planar (4) Rectangular planar

CB0177

41. The shape/structure of $[XeF_5]^-$ and XeO_3F_2, respectively, are : **[JEE-MAINS-2020]**

(1) pentagonal planar and trigonal bipyramidal

(2) trigonal bipyramidal and pentagonal planar

(3) octahedral and square pyramidal

(4) trigonal bipyramidal and trigonal bipyramidal

CB0178

42. The oxidation states of transition metal atoms in $K_2Cr_2O_7$, $KMnO_4$ and K_2FeO_4, respectively, are x, y and z. The sum of x, y and z is ____. **[JEE-MAINS-2020]**

CB0179

43. The number of CI = O bonds in perchloric acid is, "_____" **[JEE-MAINS-2020]**

CB0180

44. If the magnetic moment of a dioxygen species is 1.73 B.M, it may be : **[JEE MAINS 2020]**

(1) O_2^- or O_2^+ (2) O_2 or O_2^+ (3) O_2 or O_2^- (4) O_2, O_2^- or O_2^+

CB0183

45. Given below are two statements : **[JEE MAINS 2021]**

Statement-I : α and β forms of sulphur can change reversibly between themselves with slow heating or slow cooling.

Statement-II : At room temperature the stable crystalline form of sulphur is monoclinic sulphur.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true.

(2) Both Statement I and Statement II are true.

(3) Both Statement I is true but Statement II is false.

(4) Both Statement I and Statement II are false.

CB0184

46. Given below are two statements :one is labelled as Assertion A and the other is labelled as Reason R

[JEE MAINS 2021]

Assertion A : In $T\ell I_3$, isomorphous to CsI_3, the metal is present in +1 oxidation state.

Reason R : $T\ell$ metal has fourteen f electrons in the electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) **A** is correct but **R** is not correct

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(3) **A** is not correct but **R** is correct

(4) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.

CB0185

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47. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R. **[JEE MAINS 2021]**

Assertion A : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.

Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) A is false but R is true

(2) Both A and R are true and R is the correct explanation of A

(3) A is true R is false

(4) Both A and R are true but R is NOT the correct explanation of A

CB0186

48. Given below are two statements : **[JEE MAINS 2021]**

Statement I : None of the alkaline earth metal hydroxides dissolve in alkali.

Srtatement II : Solubility of alkaline earth metal hydroxides in water increases down the group.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) **Statement I** and **Statement II** both are incorrect.

(4) **Statement I** and **Statement II** both are correct.

CB0187

49. Match **List - I** with **List - II**. **[JEE MAINS 2022]**

List - I (Compound)	**List - II (Shape)**
(A) BrF_5	(I) bent
(B) $[CrF_6]^{3-}$	(II) square pyramidal
(C) O_3	(III) trigonal bipyramidal
(D) PCl_5	(IV) octahedral

Choose the correct answer from the options given below :

(A) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

(B) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

(C) (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

(D) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

CB0276

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50. Match **List-I** with **List-II** **[JEE MAINS 2022]**

List-I	**List-II**
(A) $\psi_{MO} = \psi_A - \psi_B$	(I) Dipole moment
(B) $\mu = Q \times r$	(II) Bonding molecular orbital
(C) $\frac{N_b - N_a}{2}$	(III) Anti-bonding molecualr orbital
(D) $\psi_{MO} = \psi_A + \psi_B$	(IV) Bond order

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III) (B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

CB0277

51. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** **[JEE MAINS 2022]**

Assertion (A) : Boron is unable to form BF_6^{3-}

Reason (R) : Size of B is very small.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(B) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(C) **(A)** is true but **(R)** is false

(D) **(A)** is false but **(R)** is true

CB0278

52. Which oxoacid of phosphorous has the highest number of oxygen atoms present in its chemical formula?

(A) Pyrophosphorous acid **[JEE MAINS 2022]**

(B) Hypophosphoric acid

(C) Phosphoric acid

(D) Pyrophosphoric acid

CB0279

53. Amongst the following the number of oxide(s) which are paramagnetic in nature is **[JEE MAINS 2022]**

Na_2O, KO_2, NO_2, N_2O, ClO_2, NO, SO_2, Cl_2O

CB0280

54. The number of interhalogens from the following having square pyramidal structure is : **[JEE MAINS 2022]**

ClF_3, IF_7, BrF_5, BrF_3, I_2Cl_6, IF_5, ClF, ClF_5

CB0281

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EXERCISE # JEE-ADVANCED

1. The molecules that will have dipole moment are : **[IIT–1992]**

(A) 2, 2–dimethylpropane (B) trans-pent-2-ene

(C) cis-hex-3-ene (D) 2, 2, 3, 3–tetramethylbutane

CB0121

2. Which of the following have identical bond order ? **[IIT–1992]**

(A) CN^- (B) O_2^- (C) NO^+ (D) CN^+

CB0122

3. Among the following the one that is polar and has the central atom with sp^2 hybridisation is :

(A) H_2CO_3 (B) SiF_4 (C) BF_3 (D) $HClO_2$ **[IIT–1997]**

CB0123

4. Which of the following is soluble in water ? **[IIT 98]**

(A) CS_2 (B) C_2H_5OH (C) CCl_4 (D) $CHCl_3$

CB0124

5. The correct order of hybridization of the central atom in the following species NH_3, $[PtCl_4]^{2-}$, PCl_5 and BCl_3 is : **[IIT 2001]**

(A) dsp^2, sp^3d, sp^2 and sp^3 (B) sp^3, dsp^2, sp^3d, sp^2

(C) dsp^2, sp^2, sp^3, sp^3d (D) dsp^2, sp^3, sp^2, sp^3d

CB0125

6. The common features among the species CN^-, CO and NO^+ are : **[IIT 2001]**

(A) bond order three and isoelectronic (B) bond order three and weak field ligands

(C) bond order two and π - acceptors (D) isoelectronic and weak field ligands

CB0126

7. Which of the following molecular specie has unpaired electron(s) ? **[JEE 2002]**

(A) N_2 (B) F_2 (C) O_2^- (D) O_2^{2-}

CB0127

8. According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ **[JEE 2004]**

(A) Paramagnetic and Bond order < O_2 (B) Paramagnetic and Bond order > O_2

(C) Diamagnetic and Bond order < O_2 (D) Diamagnetic and Bond order > O_2

CB0128

9. Among the following, the paramagnetic compound is **[JEE 2007]**

(A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2

CB0129

10. The species having bond order different from that in CO is **[JEE 2007]**

(A) NO^- (B) NO^+ (C) CN^- (D) N_2

CB0130

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11. **Statement-1 :** In water, orthoboric acid behaves as a weak monobasic acid. **[JEE 2007]**
Statement-2 : In water, orthoboric, acid acts as a proton donor.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

CB0131

12. **Statement-1 :** Pb^{+4} compounds are stronger oxidizing agents than Sn^{4+} compounds **[JEE 2008]**
Statement-2 : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to 'inert pair effect' .
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

CB0132

13. Match each of the diatomic molecules/ions in Column I with its property / properties in Column II. **[JEE 2009]**

Column I	**Column II**
(A) B_2	(P) Paramagnetic
(B) N_2	(Q) undergoes oxidation
(C) O_2^-	(R) Undergoes reduction
(D) O_2	(S) Bond order $\geq$ 2
	(T) Mixing of 's' and `p' orbitals

CB0133

14. In the reaction **[JEE 2009]**

$$2X + B_2H_6 \longrightarrow [BH_2(X)_2]^+ [BH_4]^-$$

the amine(s) X is (are)
(A) NH_3 (B) CH_3NH_2 (C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

CB0134

15. The species having pyramidal shape is **[JEE 2010]**
(A) SO_3 (B) BrF_3 (C) SiO_3^{2-} (D) OSF_2

CB0135

16. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is **[JEE 2010]**
(A) 1 and diamagnetic (B) 0 and diamagnetic (C) 1 and paramagnetic (D) 0 and paramagnetic

CB0136

Subjective

17. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is **[JEE 2010]**

CB0137

18. The total number of diprotic acids among the following is **[JEE 2010]**

H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$
H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3

CB0138

19. Among the following, the number of elements showing only one non-zero oxidation state is
O, Cl, F, N, P, Sn, Tl, Na, Ti **[JEE 2010]**

CB0139

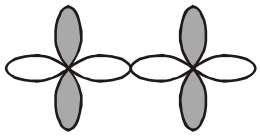
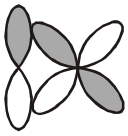
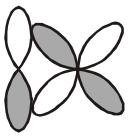
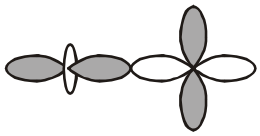
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20. Assuming 2s-2p mixing is NOT operative, the paramagnetic species among the following is : **[JEE Adv. 2014]**

(A) Be_2 (B) B_2 (C) C_2 (D) N_2

CB0140

21. Match the orbital overlap figures shown in **List-I** with the description given in **List-II** and select the correct answer using the code given below the lists. **[JEE Adv. 2014]**

List-I		**List-II**
(P)		(1) p – d π antibonding
(Q)		(2) d – d σ bonding
(R)		(3) p – d π bonding
(S)		(4) d – d σ antibonding

Code :

	P	**Q**	**R**	**S**		**P**	**Q**	**R**	**S**
(A)	2	1	3	4	(B)	4	3	1	2
(C)	2	3	1	4	(D)	4	1	3	2

CB0141

22. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is - **[JEE Adv. 2015]**

CB0142

23. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2. The **TRUE**, statement (s) regarding this adsorption is (are) **[JEE Adv. 2015]**

(A) O_2 is physisorbed

(B) heat is released

(C) occupancy of π^*_{2p} of O_2 is increased

(D) bond length of O_2 is increased

CB0143

24. According to Molecular Orbital Theory, **[JEE Adv. 2016]**

(A) C_2^{2-} is expected to be diamagnetic

(B) O_2^{2+} is expected to have a longer bond length than O_2

(C) N_2^+ and N_2^- have the same bond order

(D) He_2^+ has the same energy as two isolated He atoms

CB0144

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25. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to - **[JEE Adv. 2017]**

(A) the physical state of X_2 at room temperature changes from gas to solid down the group

(B) decrease in HOMO-LUMO gap down the group

(C) decrease in π^*-σ^* gap down the group

(D) decrease in ionization energy down the group

CB0145

26. Among H_2, He_2^+, Li_2, Be_2, B_2 ,C_2, N_2, O_2^-, and F_2 , the number of diamagnetic species is -
(Atomic number) : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8 , f = 9)

[JEE Adv. 2017]

CB0146

27. The sum of the number of lone pairs of electrons on each central atom in the following species is.

[JEE Adv. 2017]

$[TeBr_6]^{2-}$, $[BrF_2]^+$, SNF_3 and $[XeF_3]^-$

[Atomic number : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54]

CB0147

28. The option(s) with only amphoteric oxides is (are): **[JEE Adv. 2017]**

(A) Cr_2O_3, CrO, SnO, PbO

(B) NO, B_2O_3, PbO, SnO_2

(C) Cr_2O_3, BeO, SnO, SnO_2

(D) ZnO, Al_2O_3, PbO, PbO_2

CB0148

29. Among the following, the correct statement(s) is are **[JEE Adv. 2017]**

(A) $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure

(B) $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure

(C) BH_3 has the three-centre two-electron bonds in its dimeric structure

(D) The Lewis acidity of BCl_3 is greater than that of $AlCl_3$

CB0149

30. Based on the compounds of group 15 elements , the correct statement(s) is (are) **[JEE Adv. 2018]**

(A) Bi_2O_5 is more basic than N_2O_5

(B) NF_3 is more covalent than BiF_3

(C) PH_3 boils at lower temperature than NH_3

(D) The N–N single bond is stronger than the P–P single bond

CB0150

31. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE Adv. 2019]**

(1) $BeCl_2$, CO_2, BCl_3, $CHCl_3$

(2) SO_2, C_6H_5Cl, H_2Se, BrF_5

(3) BF_3, O_3, SF_6, XeF_6

(4) NO_2, NH_3, $POCl_3$, CH_3Cl

CB0151

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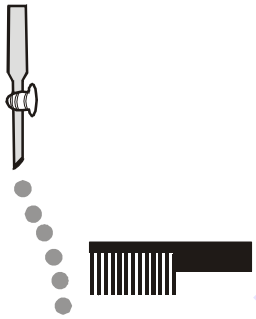
32. Among B_2H_6, $B_3N_3H_6$, N_2O, N_2O_4, $H_2S_2O_3$ and $H_2S_2O_8$, the total number of molecules containing covalent bond between two atoms of the same kind is ———— **[JEE Adv. 2019]**

CB0152

33. Consider the following compounds in the liquid form : **[JEE Adv. 2020]**

O_2, HF, H_2O, NH_3, H_2O_2, CCl_4, $CHCl_3$, C_6H_6, C_6H_5Cl.

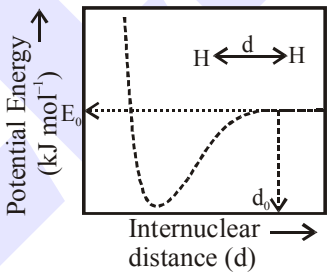
When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure?



CB0181

34. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in the figure) in kJ mol^{-1}, for $d = d_0$ at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart.

Use Avogadro constant as 6.023×10^{23} mol^{-1}. **[JEE Adv. 2020]**



CB0182

35. Which of the following liberates O_2 upon hydrolysis? **[JEE ADV. 2020]**

(A) Pb_3O_4 (B) KO_2 (C) Na_2O_2 (D) Li_2O_2

CB0188

36. For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap to two $2p_z$ orbitals is(are) **[JEE ADV. 2022]**

(A) σ orbital has a total of two nodal planes.

(B) σ^* orbital has one node in the xz-plane containing the molecular axis.

(C) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.

(D) π^* orbital has one node in the xy-plane containing the molecular axis.

CB0282

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ANSWERS KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	D	B	A	D	D	A	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	A	A	B	D	B	A	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	C	C	A	C	C	B	A	D
Que.	31	32	33							
Ans.	A	A	B							

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A, B, C	A, B, C	A, C	A, B, C	B, C	A,B,C	B,C,D	B,C,D	B,C	B, C, D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C, D	A, C	A, C	A, D	A, B, D	B, C, D	A, B, C	A, B, C	A, B, D	B, D
Que.	21	22	23	24	25	26				
Ans.	A,B,C,D	A, B	A, B, C	A, B, D	A, B, C	B,C,D				

EXERCISE # S-1

Que.	1	2	3	4	5	6	7
Ans.	8	4	2	105	4	3	4

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	A	D	A	B	B	B	D
Que.	11	12	13	14				15		
Ans.	C	B	C	(A)–Q,R,S (B)–P,Q,S (C)–S (D)–P,Q				(A)–P,S (B)–Q,T (C)–R (D)–P,S		
Que.	16					17	18	19		
Ans.	(A) – P,R,T (B) – Q (C) – Q (D) – Q,T					B	C	D		

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	3	2	3	3	3	4	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	2	4	1 & 3	4	1	4	3	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	1	1	2	2	4	4	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	2	2	2	2	2	4	4	3	1
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	1	19.00	3.00	1	3	4	1	2	C	C
Que.	51	52	53	54						
Ans.	B	D	4	3						

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B, C	A, C	A	B	B	A	C	B	D	A
Que.	11	12	13					14	15	16
Ans.	C	C	(A)-P,Q,R,T(B)-Q,R,S,T(C)-P,Q,R(D)-P,Q,R,S					B, C	D	A
Que.	17	18	19	20	21	22	23	24	25	26
Ans.	3	6	2	C	C	6	B,C,D	A,C	B, C	5 or 6
Que.	27	28	29	30	31	32	33	34		
Ans.	6	C, D	A, C, D	A, B, C	2,4	4.00	6	–2640.00 To –2620.00 or –5280.00 to –5240.00		
Que.	35	36								
Ans.	B	A, D								

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